

Week 1

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Exercise 1.0.1

A nonassociative algebra $(A, \circ)$ is called *left symmetric* if the associators are left symmetric in the sense that

$$(x, y, z) = (y, x, z) \quad ((x, y, z) := (x \circ y) \circ z - x \circ (y \circ z))$$

for all $x, y, z \in A$. Prove that for every left symmetric algebra $(A, \circ)$, $(A, [\cdot, \cdot])$ is a Lie algebra, where

$$[\cdot, \cdot] : A \times A \rightarrow A, \quad (x, y) \mapsto [x, y] := x \circ y - y \circ x$$

denotes the commutator on A .

Proof. To show $[\cdot, \cdot]$ is bilinear, we see

$$\begin{aligned} [x + y, z] &= (x + y) \circ z - z \circ (x + y) \\ &= (x \circ z - z \circ x) + (y \circ z - z \circ y) \\ &= [x, z] + [y, z] \end{aligned}$$

Similarly

$$\begin{aligned} [x, y + z] &= x \circ (y + z) - (y + z) \circ x \\ &= (x \circ y - y \circ x) + (x \circ z - z \circ x) \\ &= [x, y] + [x, z] \end{aligned}$$

Furthermore, it is obvious that

$$[x, x] = x \circ x - x \circ x = 0$$

Then we only need to show the Jacobi identity. Since $(A, \circ)$ is left symmetric,

we have

$$\begin{aligned}
& [x[yz]] + [y[zx]] + [z[xy]] \\
&= x \circ (y \circ z - z \circ y) - (y \circ z - z \circ y) \circ x \\
&+ y \circ (z \circ x - x \circ z) - (z \circ x - x \circ z) \circ y \\
&+ z \circ (x \circ y - y \circ x) - (x \circ y - y \circ x) \circ z \\
&= x \circ (y \circ z) - x \circ (z \circ y) - (y \circ z) \circ x + (z \circ y) \circ x \\
&+ y \circ (z \circ x) - y \circ (x \circ z) - (z \circ x) \circ y + (x \circ z) \circ y \\
&+ z \circ (x \circ y) - z \circ (y \circ x) - (x \circ y) \circ z + (y \circ x) \circ z \\
&= (x \circ (y \circ z) - (x \circ y) \circ z) - (x \circ (z \circ y) + (x \circ z) \circ y) \\
&- ((y \circ z) \circ x + y \circ (z \circ x)) + ((z \circ y) \circ x - z \circ (y \circ x)) \\
&- (y \circ (x \circ z) - (y \circ x) \circ z) - ((z \circ x) \circ y - z \circ (x \circ y)) \\
&= 0
\end{aligned}$$

Thus $(A, [\cdot, \cdot])$ is a Lie algebra. □

Exercise 1.0.2

Write $\text{Der}(A)$ for the Lie algebra of derivations on A , where $A = \mathbb{C}[t, t^{-1}]$. Prove that $\text{Der}(A)$ has a basis $\{L_n := t^{n+1} \frac{d}{dt} \mid n \in \mathbb{Z}\}$ and the Lie brackets among the basis elements are as follows:

$$[L_n, L_m] = (m - n)L_{m+n} \quad (m, n \in \mathbb{Z}).$$

Proof. Let $D \in \text{Der}(A)$ be any derivation on A . Then

$$\begin{aligned}
D(1) &= D(t \cdot t^{-1}) \\
&= D(t) \cdot t^{-1} + tD(t^{-1})
\end{aligned}$$

The other hand

$$D(1) = D(1 \cdot 1) = D(1) \cdot 1 + 1 \cdot D(1) = 2D(1) \implies D(1) = 0$$

Thus we have

$$D(t) \cdot t^{-1} + tD(t^{-1}) = 0 \implies D(t^{-1}) = -t^{-2}D(t)$$

From the linearity and the product rule, we observe that a derivation $D \in \text{Der}(A)$ is totally determined by its value on t , that is for any $D_1, D_2 \in \text{Der}(A)$ if $D_1(t) = D_2(t)$, then $D_1 = D_2$. It is straightforward to verify that L_n is a

derivation on A for each $n \in \mathbb{Z}$. Note that

$$L_n(t) = t^{n+1}, \quad n \in \mathbb{Z}.$$

Since $D(t) \in A$, we have

$$D(t) \in \text{span}(\dots, L_{-2}(t), L_{-1}(t), L_0(t), L_1(t), L_2(t), \dots).$$

Then we get

$$D \in \text{span}(\dots, L_{-2}, L_{-1}, L_0, L_1, L_2, \dots).$$

Suppose we have a finite linear combination of the L_n that equals the zero derivation: $\sum_{n \in I} k_n L_n = 0$, where $I \subset \mathbb{Z}$ is a finite set and $k_n \in \mathbb{C}$. To check whether this is the zero derivation, we can test its action on the generator t . $(\sum_{n \in I} k_n L_n)(t) = 0(t) = 0$. This gives: $\sum_{n \in I} k_n L_n(t) = \sum_{n \in I} k_n t^{n+1} = 0$. The set of monomials $\{t^j\}_{j \in \mathbb{Z}}$ is a basis for A as a $\mathbb{C}$ -vector space, so they are linearly independent. The equation $\sum_{n \in I} k_n t^{n+1} = 0$ implies that all coefficients must be zero, i.e., $k_n = 0$ for all $n \in I$. This proves that the set $\{L_n\}_{n \in \mathbb{Z}}$ is linearly independent. Combining above all, we know $\text{Der}(A)$ has a basis $\{L_n : n \in \mathbb{Z}\}$.

To show $[L_n, L_m] = (m - n)L_{m+n}$, we only need to show that they agree on t :

$$\begin{aligned} [L_n, L_m](t) &= (L_n(L_m(t)) - L_m(L_n(t))) \\ &= (L_n(t^{m+1}) - L_m(t^{n+1})) \\ &= ((m+1)t^{m+n+1} - (n+1)t^{m+n+1}) \\ &= (m-n)t^{m+n+1} \\ &= (m-n)L_{m+n}(t) \end{aligned}$$

which completes the proof. □

Exercise 1.0.3

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and let V be an ***n-dimensional*** vector space over $\mathbb{F}$ equipped with a nondegenerate bilinear form

$$B : V \times V \rightarrow \mathbb{F}, \quad (v, w) \mapsto B(v, w).$$

Prove that:

- (a) If $\mathbb{F} = \mathbb{C}$ and B is symmetric, then there exists a basis $\{v_1, v_2, \dots, v_n\}$

of V such that $B(v_i, v_j) = \delta_{i,j}$ for all $1 \leq i, j \leq n$.

- (b) If $\mathbb{F} = \mathbb{R}$ and B is symmetric, then there exists a basis $\{v_1, v_2, \dots, v_n\}$ of V and $p = 0, 1, \dots, n$ such that $B(v_i, v_j) = \varepsilon_i \delta_{i,j}$ for all $1 \leq i, j \leq n$, where $\varepsilon_i = 1$ if $i \leq p$ and $\varepsilon_i = -1$ if $i > p$.
- (c) If $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and B is skew-symmetric, then $n = 2m$ is even and there exists a basis $\{v_1, v_2, \dots, v_n\}$ of V such that the matrix

$$(B(v_i, v_j))_{1 \leq i, j \leq n} = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix},$$

where I_m denotes the identity matrix of rank m .

Proof.

(a)

